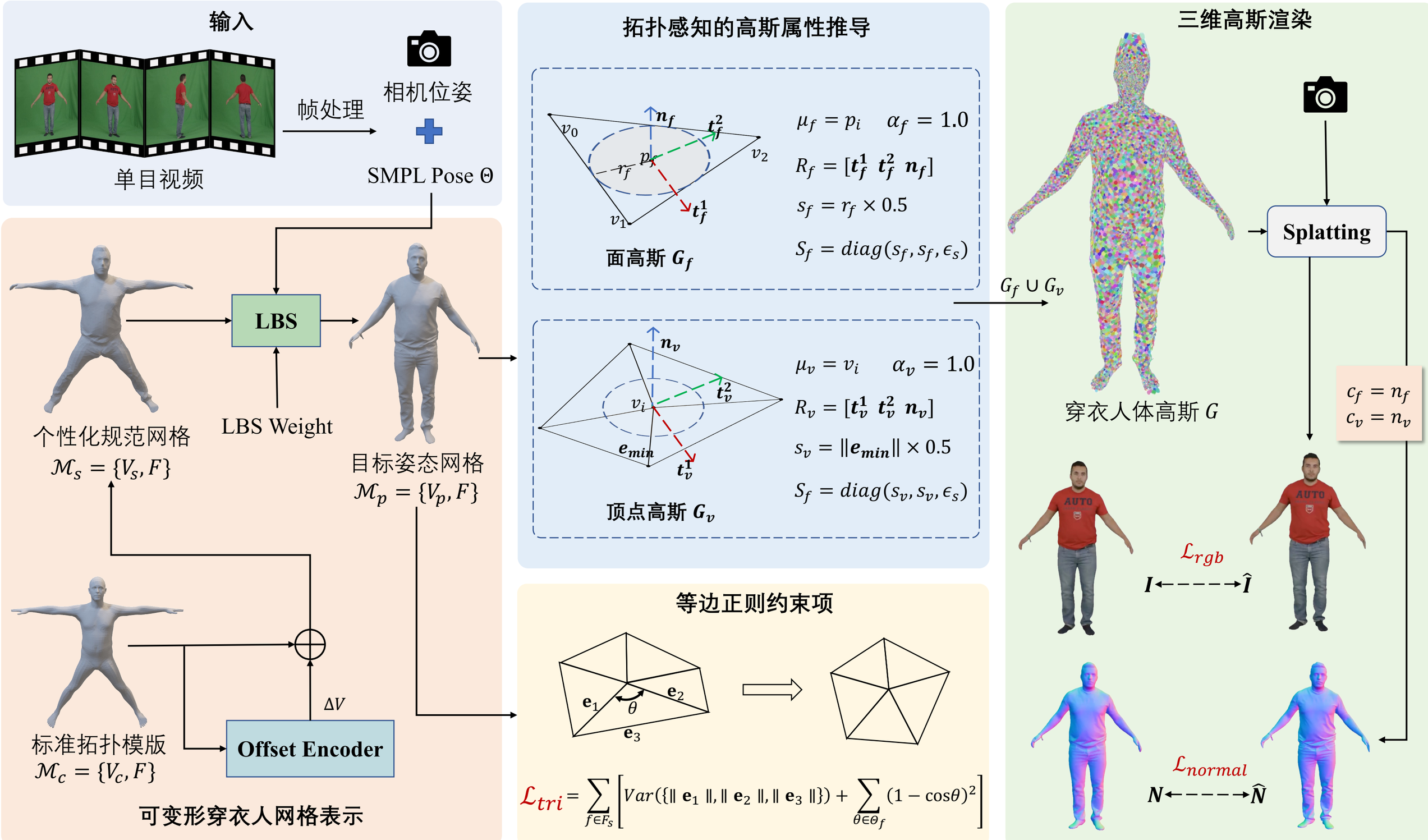
输入
单目视频
帧处理
相机位姿
SMPL Pose Θ
个性化规范网格
$\mathcal{M}_s = \{V_s, F\}$
LBS
LBS Weight
目标姿态网格
$\mathcal{M}_p = \{V_p, F\}$
标准拓扑模版
$\mathcal{M}_c = \{V_c, F\}$
ΔV
Offset Encoder
可变形穿衣人网格表示
拓扑感知的高斯属性推导
v_0 v_1 v_2 n_f t_f^2 t_f^1 p_f r_f
面高斯 G_f
$\mu_f = p_i \quad \alpha_f = 1.0$
$R_f = [t_f^1 \ t_f^2 \ n_f]$
$s_f = r_f \times 0.5$
$S_f = diag(s_f, s_f, \epsilon_s)$
n_v v_i t_v^2 e_{min} t_v^1
顶点高斯 G_v
$\mu_v = v_i \quad \alpha_v = 1.0$
$R_v = [t_v^1 \ t_v^2 \ n_v]$
$s_v = \|e_{min}\| \times 0.5$
$S_f = diag(s_v, s_v, \epsilon_s)$
等边正则约束项
e_1 e_2 e_3 θ
$\mathcal{L}_{tri} = \sum_{f \in F_s} \left[Var(\{\| e_1 \|, \| e_2 \|, \| e_3 \|\}) + \sum_{\theta \in \Theta_f} (1 - \cos\theta)^2 \right]$
三维高斯渲染
$G_f \cup G_v$
穿衣人体高斯 G
Splatting
$c_f = n_f$
$c_v = n_v$
$\mathcal{L}_{rgb}$
$I \longleftrightarrow \hat{I}$
$\mathcal{L}_{normal}$
$N \longleftrightarrow \hat{N}$